

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 1.1: INTRODUCTION TO GENERAL SCIENCE – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for Sub-strand 1.1. Use everything you have learned from the Kisumu maize-farmer puzzle, your research, the Mystery Bags activity, your Driving Question Board and your Cumulative Science Model. Answer every section in the blank space under each prompt. Write in full sentences. Use the words evidence, reasoning and inference correctly, and give Kenyan examples. Driving Question: How does a scientist know something is true when they cannot see it directly, and how is this power of General Science shaping life, careers, and the environment in Kenya?

Section 1: Explaining the Kisumu Farmer Phenomenon

Recall the maize farmer in Kisumu whose plants along one edge of the shamba were wilting and yellowing even though she watered and fertilised all the plants equally. She never saw any insect on the plants, yet her conclusion about the cause turned out to be correct. (a) List the pieces of evidence she used. (b) Describe the reasoning that connected her evidence to her conclusion. (c) State her inference, including what the hidden cause was and where it was. (d) Explain why her conclusion was not just a guess.

(a) Her evidence: the wilting and yellowing was only along one edge of the shamba, not scattered randomly; she had watered and fertilised all plants equally, so water and fertiliser were ruled out as causes; the time of season when the problem appeared; and she saw no pest above the ground. (b) Her reasoning: she combined this pattern of evidence with her knowledge of local pests. A problem that affects only one edge, and is not explained by water or fertiliser, points to a hidden local cause that damages the roots, so the plant cannot take in water and nutrients and the leaves wilt and yellow. Because the other causes were eliminated, this explained the pattern best. (c) Her inference: a soil pest was destroying the roots underground, even though she could not see it. (d) It was not a guess because it was built from several pieces of observed evidence, combined with prior knowledge and logical reasoning, and other possible causes were checked and ruled out. A guess has no evidence behind it.

Section 2: What General Science Is and Why It Matters

(a) In your own words, say what General Science is, and explain why it is more than a collection

(a) General Science is a systematic way of asking questions and finding answers using evidence. It is more than facts because it is a method: observe, collect evidence, reason, and reach conclusions that can be checked. The Kisumu farmer was practising it. (b) Human life: at Kenyatta National Hospital a doctor reads a chest X-ray and infers a hidden problem in the lung, such as an

of facts. (b) Choose TWO of the three domains: human life, environment, technology. For each, give one Kenyan example of General Science at work, and say what hidden thing the people involved had to infer or detect.	infection, from the patterns on the image; further tests would be needed to find the exact cause. Environment: NEMA monitors air quality on Mombasa Road using sensors, inferring the amount of invisible pollution from the readings. (Other acceptable examples: KEBS testing maize flour for aflatoxin, which is invisible; Safaricom engineers working with invisible radio waves for M-PESA; Nairobi Water testing for bacteria.) In each case the key action is the same as the farmer's: using indirect evidence to know about something that cannot be seen directly.
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Section 3: The Principle of Inference

(a) Define inference. (b) Name and describe the three parts of the evidence-reasoning-inference (ERI) framework. (c) Choose ONE of these Kenyan scenarios: Dr. Kamau reading a chest X-ray at Kenyatta National Hospital, or Warden Lesiamon tracking a poacher in Amboseli. Complete an ERI analysis for it.	(a) Inference is a conclusion drawn from evidence combined with prior knowledge and logical reasoning, about something that is not directly observed. It is not a guess. (b) Evidence is what can be directly observed or detected. Reasoning is the prior knowledge and logic that links the evidence to a conclusion. Inference is the conclusion about the hidden thing. (c) Example (radiologist): Evidence: the X-ray shows a cloudy white patch in one part of the patient's lung. Reasoning: from medical knowledge, lung tissue that is infected or filled with fluid appears cloudy on an X-ray, and healthy lung does not. Inference: there is probably an infection or fluid in that part of the lung, which cannot be seen directly; other tests would be needed to find the exact cause. (Example, ranger: Evidence: broken grass, footprints and discarded items along a path. Reasoning: animals do not leave those marks together and the pattern follows a route. Inference: a poacher passed this way.)
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Section 4: Collecting Evidence and Process Skills

In the Mystery Bags activity you used written clue cards to work out a hidden cause on a Kenyan farm. (a) Explain the difference between an observation and an inference, with an example. (b) Name at least three science process skills your group used and say how you used each one. (c) What should a scientist do when one clue contradicts the others, and why?	(a) An observation is what you directly notice or record, for example 'the leaves are yellow with curled edges.' An inference is a conclusion about the cause that you work out from observations, for example 'the plant has a nutrient deficiency.' (b) Classifying: we sorted clues into categories of pest, disease and nutrient or environmental problem using the Evidence Classification Table. Hypothesising: we proposed a possible cause to explain the clues. Identifying variables: we noted factors such as soil, water and season that could change the result. Interpreting data: we looked for patterns in the clues. Drawing conclusions: we stated our inference using the frame 'Based on [evidence] and [evidence], our group infers [cause] because [reasoning].' (c) The scientist should weigh the contradicting clue and not ignore it. It may show that the first idea is wrong, or it may be explained by another factor. An inference is only reliable if it accounts for all the evidence.
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Section 5: Careers, a New Scenario and Your Growth

A tea farmer in Kericho notices that the leaves on some of her tea bushes are curling. She waters and cares for all her	(a) Evidence: the curled leaves are mostly in one part of the field while the other bushes, given the same care, are healthy; and the curling began during the dry season. (b) Inference (one reasonable example): something hidden in the soil or at the roots of the affected bushes, such as a soil pest or a soil problem, is affecting them. Reasoning: all the bushes get the same care, so care alone does not explain why only one part is affected; a hidden, local cause fits a pattern confined to one area, as with the Kisumu
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bushes in the same way. The curling leaves are mostly on the bushes in one part of the field, the other bushes look healthy, and the curling began during the dry season. She cannot see any pest or disease on the leaves. (a) Identify two pieces of evidence from this scenario. (b) State one reasonable inference about what might be causing the curling, and explain your reasoning from the evidence. (c) Name a General Science career in Kenya that would investigate this professionally and say how that person would use inference. (d) Finally, answer the Driving Question in 3 to 4 sentences, explaining how your thinking has changed since Lesson 1.	maize farmer, and dry-season timing suggests stress on the roots. Other inferences that use the evidence and reasoning well, such as a disease or a local soil condition, are also acceptable. More evidence, such as examining the roots and testing the soil, would be needed to confirm it. (c) A scientist at the Tea Research Institute in Kericho, or an agronomist, would test the soil, examine the roots and compare the evidence to known pest and disease patterns, inferring the hidden cause from indirect evidence and then checking it. (d) A scientist knows something is true when they cannot see it by collecting evidence, using reasoning based on prior knowledge, and drawing an inference, as the Kisumu farmer did. This power is shaping life, careers and the environment in Kenya: doctors at KNH, food-safety officers at KEBS, and NEMA officers all use it. At the start I thought you must see something to believe it; now I understand that careful evidence and reasoning can give reliable conclusions about hidden things.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific evidence from the Kisumu farmer, the Kenyan examples and the Mystery Bags activity accurately and links each to the answer.	Uses some evidence from the lessons correctly, but links to the answer are sometimes unclear.	Gives few or vague examples, or uses evidence that is incorrect or not from the lessons.
Understanding of inference	Defines inference correctly, explains evidence, reasoning and inference clearly, and explains why inference is not a guess.	Defines inference and names the three parts, with limited explanation of how they connect.	Confuses inference with guessing or observation, or cannot name the parts.
Application to a new scenario	Applies the ERI framework to the Kericho tea scenario using the evidence given, a logical inference about a hidden cause that follows from that evidence, and a suitable career.	Applies the framework to the new scenario but the reasoning or career link is incomplete.	Gives an inference without evidence or reasoning, or does not name a career.

Science process skills and Kenyan contexts	Correctly names and explains at least three process skills and the difference between observation and inference, with relevant Kenyan contexts throughout.	Names two or three process skills with some explanation and includes some Kenyan examples.	Names few or no process skills, or examples are missing or not relevant.
Answer to the Driving Question and communication	Gives a clear, complete answer to the Driving Question that shows how thinking has changed, written in clear, organised sentences.	Answers the Driving Question partly, with some reflection on change in thinking.	Answer to the Driving Question is incomplete or unclear, with no reflection.

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